

BUCKLEY AFB, CO, USA

WMO: 724695

Lat: 39.717N

Lon: 104.750W

Elev: 1726

StdP: 82.24

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23036

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-16.9	-13.3	-20.2	0.8	-6.7	-18.0	0.9	-4.2	11.9	2.6	10.7	4.2	2.9	170	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	34.1	14.7	32.7	14.8	31.3	14.7	18.0	25.6	17.3	25.5	16.6	25.1	3.8	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	14.2	19.0	15.1	13.3	19.0	14.1	12.4	18.9	57.4	25.2	55.1	25.2	53.0	25.1	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.2	8.1	-20.7	37.0	2.6	1.2	-22.6	37.9	-24.2	38.6	-25.7	39.3	-27.6	40.2		
(5)			-20.9	19.8	2.6	0.8	-22.8	20.4	-24.3	20.9	-25.8	21.3	-27.7	21.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	0.7	1.4	5.5	8.7	13.7	19.7	23.4	22.2	18.0	10.8	4.5	0.9		
	DBStd	9.59	6.01	6.38	5.90	5.27	4.79	3.90	2.78	2.69	4.77	5.57	6.28	5.78		
	HDD10.0	1327	290	243	159	85	21	1	0	0	7	57	181	283		
	HDD18.3	3149	546	473	397	289	155	29	2	4	61	237	416	540		
	CDD10.0	1634	2	3	20	46	135	291	416	377	247	81	16	2		
	CDD18.3	416	0	0	0	0	11	69	159	124	51	2	0	0		
	CDH23.3	4913	0	1	5	19	213	926	1761	1257	651	79	1	0		
	CDH26.7	2078	0	0	0	1	53	391	857	534	232	9	0	0		
(14) Wind	WSAvg	4.0	3.9	4.1	4.5	4.7	4.2	3.9	3.8	3.7	3.6	3.7	3.8	3.9		
(15) Precipitation	PrecAvg	428	14	18	32	55	58	48	55	54	36	28	21	15		
	PrecMax	613	32	56	108	140	149	111	111	115	174	71	68	71		
	PrecMin	248	2	0	0	3	15	12	8	8	0	0	6	0		
	PrecStd	86	8	13	26	30	33	27	30	27	33	19	15	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.6	20.0	23.9	25.3	31.3	35.0	36.3	35.0	33.6	27.8	22.7	17.9		
		MCWB	6.4	6.6	8.4	9.7	13.5	14.2	15.0	14.8	14.3	11.1	7.8	5.7		
	2%	DB	15.0	17.3	21.2	23.4	27.8	32.8	34.8	33.0	31.2	25.8	20.0	14.9		
		MCWB	4.8	5.5	7.6	9.1	12.4	14.1	15.0	15.0	13.4	10.3	7.4	4.5		
	5%	DB	12.3	14.4	18.6	21.6	25.9	31.2	33.0	31.7	29.6	23.3	17.4	12.3		
		MCWB	3.5	4.4	6.6	8.5	11.9	14.1	15.1	15.0	13.2	9.6	6.5	3.4		
	10%	DB	9.7	11.6	15.9	18.9	23.0	28.9	31.9	30.0	27.6	20.5	14.1	9.4		
		MCWB	2.4	3.4	5.5	7.7	11.3	14.2	15.3	15.2	13.0	9.0	5.0	2.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	7.3	9.1	11.3	16.0	18.4	19.2	18.5	17.0	13.5	8.9	6.3		
		MCDB	17.3	18.3	21.4	21.0	22.1	26.6	25.6	25.7	25.2	22.6	19.8	16.9		
	2%	WB	5.1	5.8	8.0	10.1	14.4	16.9	18.2	17.6	15.6	11.7	7.7	4.8		
		MCDB	13.6	16.0	19.5	20.5	23.0	24.8	25.4	26.3	24.7	20.2	18.8	14.1		
	5%	WB	3.9	4.8	6.9	9.1	13.2	16.1	17.6	16.9	14.8	10.6	6.7	3.6		
		MCDB	11.5	13.8	17.0	19.0	22.2	24.3	25.4	25.6	24.3	20.3	16.6	11.6		
	10%	WB	2.6	3.6	5.8	8.2	12.2	15.3	16.9	16.3	14.0	9.7	5.4	2.3		
		MCDB	9.1	11.1	14.9	17.4	21.1	23.9	25.2	24.9	23.5	19.2	13.9	9.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	13.0	14.0	13.6	13.5	14.9	15.0	14.5	15.0	14.2	13.1	12.7		
		MCDBR	15.7	16.7	17.5	17.6	17.0	17.8	17.1	16.9	17.3	17.7	17.0	15.7		
		MCWBR	8.3	8.4	8.0	7.3	6.4	5.7	4.7	4.8	5.4	7.1	8.1	8.2		
	5% WB	MCDBR	15.2	15.9	16.5	16.1	15.3	14.6	14.3	14.1	14.9	15.5	16.3	15.2		
		MCWBR	8.4	8.4	7.9	7.2	6.4	6.0	5.0	4.8	5.3	6.7	7.9	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.236	0.243	0.263	0.294	0.313	0.321	0.349	0.339	0.304	0.266	0.248	0.237		
	taud		2.631	2.625	2.571	2.492	2.501	2.490	2.452	2.467	2.546	2.642	2.613	2.619		
	Ebn at Noon		972	1005	1006	983	964	951	921	924	945	961	944	943		
	Edh at Noon		65	76	90	104	106	107	111	106	91	74	66	61		
(44) All-Sky Solar Radiation	RadAvg		2.57	3.55	4.76	5.68	6.30	7.18	6.86	6.14	5.28	3.88	2.79	2.25		
	RadStd		0.11	0.19	0.35	0.39	0.49	0.33	0.30	0.20	0.27	0.24	0.15	0.12		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	+0.30	+0.68	N/A	N/A	N/A	N/A	
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+58	N/A	

Nomenclature: See separate page